

Gaussian Limit of the Hyperbolic Gravitational-Wave Likelihood

Setup

For one detector and one frequency bin, define the complex residual

$$r_k = d_k - h_k, \quad (1)$$

and write the Bilby per-bin scale as

$$s_k^2 = \frac{S_n(f_k) T}{4}, \quad (2)$$

where $S_n(f_k)$ is the one-sided noise power spectral density and T is the data duration.

The hyperbolic likelihood used in the implementation treats the real and imaginary parts of r_k as a two-dimensional real vector. Define the standardized residual

$$x_k = \frac{r_k}{\sqrt{s_k^2}}, \quad q_k = |x_k|^2 = \frac{|r_k|^2}{s_k^2}. \quad (3)$$

The per-bin symmetric hyperbolic density is

$$p(x_k | \alpha, \delta) = \frac{\alpha}{2\pi\delta K_1(\alpha\delta)} K_0\left(\alpha\sqrt{\delta^2 + q_k}\right), \quad (4)$$

with shape parameters $\alpha > 0$ and $\delta > 0$, and K_ν the modified Bessel function of the second kind.

Asymptotic Limit

Take the limit

$$\alpha, \delta \rightarrow \infty, \quad \lambda = \frac{\alpha}{\delta} \text{ fixed.} \quad (5)$$

Use the large-argument asymptotic expansion

$$K_\nu(z) \sim \sqrt{\frac{\pi}{2z}} e^{-z} \quad (z \rightarrow \infty). \quad (6)$$

Applying this to both Bessel functions gives

$$p(x_k | \alpha, \delta) \sim \frac{\alpha}{2\pi\delta} \frac{\sqrt{\frac{\pi}{2\alpha\sqrt{\delta^2 + q_k}}} e^{-\alpha\sqrt{\delta^2 + q_k}}}{\sqrt{\frac{\pi}{2\alpha\delta}} e^{-\alpha\delta}} \quad (7)$$

$$= \frac{\alpha}{2\pi\delta} \sqrt{\frac{\delta}{\sqrt{\delta^2 + q_k}}} \exp\left[-\alpha\left(\sqrt{\delta^2 + q_k} - \delta\right)\right]. \quad (8)$$

Now expand the square root:

$$\sqrt{\delta^2 + q_k} = \delta \sqrt{1 + \frac{q_k}{\delta^2}} = \delta + \frac{q_k}{2\delta} + O(\delta^{-3}). \quad (9)$$

Therefore

$$\alpha \left(\sqrt{\delta^2 + q_k} - \delta \right) = \frac{\alpha}{2\delta} q_k + O(\delta^{-2}) \rightarrow \frac{\lambda q_k}{2}, \quad (10)$$

and also

$$\sqrt{\frac{\delta}{\sqrt{\delta^2 + q_k}}} \rightarrow 1. \quad (11)$$

Hence the limiting density is

$$p(x_k | \alpha, \delta) \rightarrow \frac{\lambda}{2\pi} \exp\left(-\frac{\lambda q_k}{2}\right). \quad (12)$$

This is the density of a two-dimensional zero-mean Gaussian with precision λ .

Limit in Terms of the Residual r_k

Since $q_k = |r_k|^2 / s_k^2$, the density in the original residual variable is

$$p(r_k) \rightarrow \frac{\lambda}{2\pi s_k^2} \exp\left(-\frac{\lambda |r_k|^2}{2s_k^2}\right). \quad (13)$$

Equivalently, the per-bin log-likelihood becomes

$$\log p(r_k) \rightarrow \log \lambda - \log(2\pi s_k^2) - \frac{\lambda |r_k|^2}{2s_k^2}. \quad (14)$$

For the special case $\alpha = \delta \rightarrow \infty$, one has $\lambda = 1$, so

$$\log p(r_k) \rightarrow -\log(2\pi s_k^2) - \frac{|r_k|^2}{2s_k^2}. \quad (15)$$

Substituting the Bilby scale

$$s_k^2 = \frac{S_n(f_k) T}{4}, \quad (16)$$

gives

$$\log p(r_k) \rightarrow -\log\left(\frac{\pi S_n(f_k) T}{2}\right) - \frac{2|r_k|^2}{T S_n(f_k)}. \quad (17)$$

This is exactly the standard complex Gaussian per-bin likelihood used by Bilby. Summing over independent frequency bins and detectors reproduces the usual Gaussian/Whittle gravitational-wave likelihood, up to the same additive normalization constants already present in the Gaussian model.

Conclusion

The symmetric hyperbolic likelihood has a Gaussian limit only when

$$\alpha, \delta \rightarrow \infty \quad \text{with} \quad \lambda = \frac{\alpha}{\delta} \quad \text{fixed}. \quad (18)$$

The limiting Gaussian precision is λ . In particular, choosing $\alpha = \delta$ and taking both to infinity yields the standard Bilby Gaussian likelihood.